

Solar Site Suitability Assessment

LOCATION: Rosamond Corridor E (Solar Star core) · NERC REGION: WECC

SITE SCORE

86 / 100

High

CONFIDENCE

95 / 100

HIGH

"This assessment is based on authoritative data for all major criteria."

DATA GAPS

- solar_ej: No data available for Environmental Justice. Tried: epa_ejscreen

INTERCONNECTION CONTEXT

Nearest substation: (unnamed substation) (2.1 mi, 220 kV)

Existing capacity (within 50 km): 4375.9 MW across 141 operating plants (EIA Form 860).

Queue activity (n/a): 0 active solar interconnection requests totaling 0 MW within 50 km.

NOTE: Informational context from public data, not an interconnection study. Actual capacity availability requires an interconnection application with the ISO/RTO. Queue data: LBNL "Queued Up" 2025, county-centroid precision.

EXCLUSION SCREENING

Critical habitat	✓ pass	usfws_critical_habitat
Protected areas	✓ pass	usgs_padus
Steep slope	✓ pass	usgs_3dep
Urban land	✓ pass	nlcd_land_cover
Wetlands	✓ pass	nwi_wetlands

SCORED CRITERIA

solar_transmission	0.42	91	hifld_transmission	HIGH
solar_ghi	0.10	93	nrel_pvwatts_v8	HIGH
solar_slope	0.17	95	usgs_3dep	HIGH
solar_road	0.10	50	osm_roads_overpass	HIGH
solar_aspect	0.08	100	usgs_3dep	HIGH
solar_land_cover	0.08	60	nlcd_land_cover	HIGH

solar_soil	0.05	100	usda_sda_ssurgo	HIGH
excl_flood	0.05	50	fema_nfhl	HIGH
solar_ej	0.00	—	—	NONE

WEIGHT PROFILE: WECC — calibrated against 100 EIA Form 860 installations

Methodology

FRAMEWORK

GIS-based Multi-Criteria Decision Analysis (GIS-MCDA)

- Doorga et al. (2019) — Renewable and Sustainable Energy Reviews 104:133-146
- Hernandez et al. (2015) — PNAS 112:13579-13584
- Al-Shammari et al. (2026) — Renewable Energy (in press)
- Charabi & Gastli (2011) — Renewable Energy 36(9):2554-2561
- Ong et al. (2013) — NREL/TP-6A20-56290

WEIGHT CALIBRATION

Constrained optimization (scipy SLSQP) within literature-supported bounds against EIA Form 860 ground truth, per NERC region.

DATA SOURCES

14 criteria (9 scored, 5 exclusion) from federal and open data sources, with per-criterion source selection via the data-selection engine and quality-ordered data trees.

KNOWN LIMITATIONS

- NWI wetlands data unavailable nationally (SSURGO hydric-soil proxy used).
- EJScreen data is a static 2024 snapshot (EPA tool discontinued).
- Scoring does not assess interconnection capacity or queue position.

DISCLAIMER: This assessment is based on publicly available federal data and peer-reviewed methodology. It is intended for screening purposes and does not replace site-specific field investigation, interconnection studies, or environmental surveys.